

SEO PACKAGE

1. SEO Title

SQL ORDER BY Explained: How to Sort Query Results (ASC, DESC, NULLs, and Multiple Columns)

2. SEO Subtitle

Learn why databases return rows in unpredictable order and how `ORDER BY` gives you full, repeatable control over your query results.

3. Featured Quote

"A `SELECT` without `ORDER BY` isn't sorted wrong — it isn't sorted at all. The database never promised you an order in the first place."

4. Meta Description

Learn how SQL `ORDER BY` sorts query results with `ASC`, `DESC`, multiple columns, and expressions. Understand `NULL` placement and avoid common beginner mistakes.

5. URL Slug

sql-order-by-sort-query-results

6. Keywords

SQL `ORDER BY`, sort SQL results, `ASC` `DESC` SQL, SQL sorting, order by multiple columns, SQL `NULL` sorting, `ORDER BY` vs `GROUP BY`, SQL query ordering, sort database results, SQL for beginners, learn SQL, database sorting, SQL `SELECT` order, sort by expression SQL, SQL tutorial, coding interview SQL, data analysis SQL, SQL ascending descending, deterministic ordering SQL, SQL report sorting

7. Tags

SQL , ORDER BY , Database , SQL Tutorial , Data Analysis , Learn to Code , Backend , SQL for Beginners , Query Optimization , Software Engineering , Coding Interview , Data Engineering , RDBMS , SQL Basics

SQL ORDER BY: How to Take Control of Your Query Result Order

8. Introduction

Here's a moment that trips up almost every developer learning SQL. You write a `SELECT` query, run it, and the rows come back looking perfectly sorted. You ship it. Two weeks later, the same query returns the same rows — but in a completely different order — and a report that looked clean is now a jumbled mess.

Nothing broke. You just relied on something the database never promised.

Understanding result ordering is one of the first real "aha" moments in learning SQL, and it separates people who *use* databases from people who *understand* them. Interviewers love this topic because it reveals whether you grasp a core truth about relational databases: **a table is a set of rows, and sets have no inherent order.** The moment you accept that, `ORDER BY` stops feeling like a formatting afterthought and starts feeling like the tool that gives you determinism, repeatability, and control.

Whether you're building a leaderboard, a paginated API, a financial report, or a sequential data pipeline, ordering is not optional. This lesson walks you through *why* rows come back scrambled, *how* `ORDER BY` fixes it, and the sharp edges — like `NULL` placement — that quietly cause production bugs.

9. Problem Overview

When you run a `SELECT` statement, the database returns a **result set**: the collection of rows that matched your query. Think of it like a printed receipt of records.

The critical, counterintuitive rule is this: **unless you explicitly ask for an order, the database is free to return those rows in any order it likes.** It will typically pick whatever is fastest — whatever matches how the data physically sits on disk, how the query planner walked an index, or how

parallel workers happened to finish. That "order" can change between runs, between database versions, and between servers.

Our task is simple to state: **take an unordered result set and arrange it into a specific, repeatable order** — smallest-to-largest, newest-to-oldest, alphabetically, or by some value we compute on the fly. That's exactly what the `ORDER BY` clause is for.

10. Example

Imagine a `students` table:

id name class score

1	Ada	B	88
2	Ben	A	60
3	Chen	A	95
4	Dara	B	88
5	Evan	A	72

Run `SELECT name, score FROM students;` and you might get Ada, Ben, Chen, Dara, Evan — or Chen, Evan, Ben, Dara, Ada. **Both are valid.** The database owes you no particular order.

Now run:

```
SELECT name, score
FROM students
ORDER BY score DESC;
```

Output:

name score

Chen	95
Ada	88
Dara	88
Evan	72
Ben	60

Every single time. The `95` slides to the front, the `60` drops to the back. No data changed — we only rearranged the same five rows into a sensible line. (Notice Ada and Dara both scored 88; we'll fix that tie in a moment.)

11. Intuition

Picture a deck of cards dumped face-up on a table. That pile is your unsorted result set — technically all the cards are "there," but there's no meaningful sequence. If you want them in order, you have to *hand the deck to someone and say how to sort it*.

That's `ORDER BY`. You name a column, and you name a direction, and the database lines the rows up accordingly.

The mental model that makes everything click: **the database sorts on the *value* in the column you name, not on the row's position**. So you can sort by a number (small to big), by text (alphabetical), by a date (oldest to newest), or even by a value you calculate. And when one column produces ties, you hand over a *second* card-sorting rule to break them. Once you see sorting as "give the database a comparison rule," multi-column sorting and expression sorting stop being separate features — they're the same idea with a richer rule.

12. Brute Force Solution

Idea: Skip `ORDER BY` entirely. Pull the rows into your application, then sort them in your own code (Python, JavaScript, whatever).

```
rows = db.execute("SELECT name, score FROM students")
rows_sorted = sorted(rows, key=lambda r: r["score"], reverse=True)
```

Algorithm: 1. Fetch every matching row from the database. 2. Load them all into memory in the app. 3. Run a sort in application code.

Advantages: - Works when your ordering rule is genuinely too weird for SQL to express (custom business logic, locale rules a specific engine can't handle). - Full control in a language you may know better than SQL.

Disadvantages: - You must transfer *all* rows over the network before sorting — wasteful, especially with `LIMIT`-style "top 10" needs. - You reimplement something the database already does better. - The database can't use indexes to help you. - More code, more surface area for bugs.

Time complexity: $O(n \log n)$ for the sort, plus the cost of moving all n rows to the app. **Space complexity:** $O(n)$ — the entire result set must sit in application memory.

13. Optimal Solution

Let the database do the sorting with `ORDER BY`. It's purpose-built for this, can lean on indexes, and it's one clean line.

Step 1 — Sort by a single column.

```
SELECT name, score
FROM students
ORDER BY score;      -- ascending by default
```

Step 2 — Choose a direction. `ASC` (ascending, small → big) is the default; `DESC` (descending, big → small) is the one you must request explicitly.

```
ORDER BY score DESC; -- leaderboard: highest first
```

Keyword Direction Default?

`ASC` small → big ☒ yes

`DESC` big → small ☐ ask for it

Step 3 — Break ties with multiple columns. SQL reads the list **left to right**: it sorts by the first column, and only reaches for the next column when two rows tie.

```
SELECT name, class, score
FROM students
ORDER BY class ASC, name ASC;
```

This sorts by `class` first, then alphabetically by `name` *within* each class:

name class score

Ben	A	60
Chen	A	95
Evan	A	72
Ada	B	88
Dara	B	88

`class` is the primary rule; `name` is the tie-breaker. **List your columns in order of importance.**

Step 4 — Sort by a computed expression. You aren't limited to plain columns. The database evaluates the expression first, then sorts on the result.

```
SELECT product, price, quantity
FROM orders
ORDER BY price * quantity DESC; -- biggest total orders on top
```

Step 5 — Know where NULLs land. A `NULL` is a blank — a value nobody filled in. When you sort, all the NULLs clump together at one end. **Which end depends on your database:** PostgreSQL

puts NULLs *last* in `ASC` ; MySQL puts them *first*. Don't assume — control it explicitly where supported:

```
ORDER BY score DESC NULLS LAST;
```

14. Python Code

Here's a small, self-checking script that shows both the raw (undefined-order) result and the correctly sorted one using SQLite from the standard library.

```
import sqlite3

def build_db() -> sqlite3.Connection:
    """Create an in-memory students table and populate it."""
    conn = sqlite3.connect(":memory:")
    conn.execute(
        "CREATE TABLE students (id INTEGER, name TEXT, class TEXT, score INTEGER)"
    )
    conn.executemany(
        "INSERT INTO students VALUES (?, ?, ?, ?)",
        [
            (1, "Ada", "B", 88),
            (2, "Ben", "A", 60),
            (3, "Chen", "A", 95),
            (4, "Dara", "B", 88),
            (5, "Evan", "A", 72),
        ],
    )
    return conn

def top_scores(conn: sqlite3.Connection) -> list[tuple[str, int]]:
    """Return (name, score) sorted highest score first, deterministically."""
    cursor = conn.execute(
        """
        SELECT name, score
        FROM students
        ORDER BY score DESC, name ASC    -- name breaks score ties
        """
    )
    return cursor.fetchall()

def demo() -> None:
    conn = build_db()
    result = top_scores(conn)
```

```
# The tie between Ada and Dara (both 88) is broken by name ASC.
assert result == [
    ("Chen", 95),
    ("Ada", 88),
    ("Dara", 88),
    ("Evan", 72),
    ("Ben", 60),
], result
print("Sorted result:", result)
print("All checks passed.")

if __name__ == "__main__":
    demo()
```

15. Code Walkthrough

- `build_db()` spins up an in-memory SQLite database so the example runs anywhere with no setup. It creates the table and inserts our five students.
- `top_scores()` is the heart of it. The query orders by `score DESC` (highest first) and adds `name ASC` as a tie-breaker. Without that second column, the tie between Ada and Dara (both 88) would resolve in an *undefined* order — the exact bug this lesson warns about.
- `ORDER BY score DESC, name ASC` — the comma-separated list is read left to right. `score` is primary; `name` only matters when scores are equal.
- `demo()` runs the query and asserts the *exact* expected order. This is what turns "it looked right" into "it is provably right and repeatable." If the ordering logic ever breaks, the `assert` fails loudly.

16. Dry Run

Let's trace `ORDER BY score DESC, name ASC` over the data.

1. Start with the five rows in arbitrary order.
2. Apply the primary rule, `score DESC` : $95 \rightarrow 88 \rightarrow 88 \rightarrow 72 \rightarrow 60$.
3. Chen (95), then the two 88s (Ada, Dara), then Evan (72), then Ben (60).
4. Two rows tie at 88. Apply the tie-breaker, `name ASC` : "Ada" < "Dara" alphabetically, so Ada comes first.
5. Final order locked in:

Step name score

1 Chen 95

Step name score

```

2   Ada   88
3   Dara  88
4   Evan  72
5   Ben   60

```

Run it a thousand times — identical output every time. That's determinism.

17. Complexity Analysis

Sorting is fundamentally a comparison operation, so:

- **Time:** $O(n \log n)$ in the general case, where n is the number of rows in the result set. This is the lower bound for comparison sorts, and it's why sorting large result sets isn't free.
- **Space:** $O(n)$ when the database must materialize and sort the rows in memory (or spill to disk if the set is huge).

The crucial optimization: **if an index already exists on the ORDER BY column(s) in the right direction, the database can read rows in sorted order directly and skip the sort entirely** — effectively $O(1)$ extra sorting work. This is the single biggest reason to sort in the database rather than the app: the database can *see* those indexes; your application code cannot.

18. Alternative Solutions

Sort in the application layer (the brute-force approach from Section 12). Fetch rows, sort in Python/JS. Reasonable only when the ordering rule can't be expressed in SQL — for example, a bespoke ranking that depends on live external data. For everything else, in-database sorting wins on network cost, index usage, and code simplicity.

Comparison at a glance:

Approach	Network cost	Uses indexes	Code size	Best for
ORDER BY (database)	Low	✓	One line	Almost everything
App-side sort	High (all rows)	✗	More logic	Rules SQL can't express

19. Edge Cases

- **Empty result set:** ORDER BY on zero rows is perfectly valid — you get zero rows back, no error.

- **NULL values:** They clump at one end, and *which* end is engine-dependent (Postgres: last in ASC; MySQL: first). Use `NULLS FIRST` / `NULLS LAST` where supported.
- **Ties with no tie-breaker:** Rows with equal sort keys come back in undefined relative order. Add a unique tie-breaker column (like `id`) for full determinism.
- **Mixed case text:** `'Ada'` vs `'ada'` may sort differently depending on the column's collation. Know your collation.
- **Very large sets:** Sorting can spill to disk and slow down. An index or a `LIMIT` helps enormously.

20. Common Mistakes

1. **Assuming rows are sorted because they were last time.** The single most common trap. Without `ORDER BY`, order is never guaranteed.
2. **Forgetting `DESC`.** You want the *highest* scores but write plain `ORDER BY score`, so you get the lowest instead. `ASC` is the silent default.
3. **Assuming `NULL` placement.** People expect `NULLs` at the bottom and get a wall of blanks at the top. Always verify, or specify `NULLS LAST`.
4. **Confusing `ORDER BY` with `GROUP BY`.** They rhyme, but `ORDER BY` *arranges* rows while `GROUP BY` *collapses* them into summary buckets.
5. **No tie-breaker in paginated queries.** If your sort key isn't unique, pagination can show or skip rows unpredictably as the undefined tie order shifts between page fetches. Always end `ORDER BY` with a unique column.

21. Interview Questions

- "Why might the same `SELECT` return rows in a different order on two runs?" — Because a result set has no inherent order without `ORDER BY`.
- "How does SQL resolve ties when sorting?" — It moves to the next column in the `ORDER BY` list; if none exists, order is undefined.
- "Where do `NULLs` sort, and how do you control it?" — Engine-dependent; use `NULLS FIRST` / `NULLS LAST`.
- "When would you sort in the app instead of the database?" — Only when the ordering rule can't be expressed in SQL.
- "Why is `ORDER BY` important for pagination?" — Stable, unique ordering keeps `LIMIT` / `OFFSET` consistent across pages.

22. Similar Problems

- **LeetCode 595 – Big Countries:** filtering plus ordering output for a report.
- **LeetCode 1741 – Find Total Time Spent by Each Employee:** combines `GROUP BY` aggregation with an `ORDER BY` on the result — the exact "sort *and* summarize" pattern.
- **LeetCode 176 – Second Highest Salary:** ranking by value, where understanding sort order and ties is essential.

These reinforce that ordering rarely stands alone; it pairs with filtering, grouping, and ranking in real queries.

23. Key Takeaways

- A result set has **no inherent order** — `ORDER BY` is how you impose one.
- `ASC` is the default; **`DESC` is the one you must ask for.**
- List multiple columns **in order of importance** to break ties.
- You can sort by columns, text, dates, or **computed expressions**.
- **Verify where NULLs land** — it's engine-dependent.
- `ORDER BY` sorts; `GROUP BY` summarizes. They're different tools.
- **Sort in the database** unless the rule is too unusual for SQL.

24. Watch the Video

The full walkthrough — with visual card-deck demos of exactly how rows rearrange, live ASC vs. DESC comparisons, and the NULL "gotcha" that burned me once in a real report — is in the video. Watching it makes the intuition stick far faster than reading alone:

👉 [Watch SQL Lesson 4 here](#)

25. About the Series

This is Lesson 4 of the **Fun with Learning Technology** series — a beginner-friendly journey through SQL, one focused concept at a time. Each lesson builds real intuition first, then the syntax, so you actually understand *why* things work instead of memorizing commands. No jargon walls, no assumed background — just clear, practical database skills you'll reach for on the job.

26. Call To Action

If this cleared up something that used to confuse you:

- 👍 **Like** the video on YouTube — it genuinely helps the series reach more learners.
- 🔔 **Subscribe** so you catch every SQL lesson.
- 📺 **Subscribe on Substack** for the written deep-dives like this one.
- 💬 **Comment** with the SQL topic you want next.
- 💻 **Share** it with a friend who's learning to code.

This problem had a trick. Liking the video doesn't need one.

SOCIAL MEDIA

LinkedIn Post

Early in your SQL journey, there's a bug that isn't a bug — and it confuses almost everyone.

You write a `SELECT`, the rows come back looking neatly ordered, and you move on. Weeks later the *same query* returns the *same rows* in a *different order*, and a report that looked perfect is suddenly a mess.

Nothing broke. You relied on something the database never promised.

Here's the core truth: a table is a set of rows, and a set has no inherent order. Without an explicit instruction, the database returns rows in whatever order was fastest — matching physical layout, index walks, or parallel workers finishing. That order can change between runs and servers.

The fix is `ORDER BY`:

→ Name the column to sort on. → `ASC` (small → big) is the default; `DESC` is the one you must request. → List multiple columns to break ties — read left to right, in order of importance. → You can even sort by computed expressions like `price * quantity`.

And the sharp edge that catches everyone: NULLs. They clump at one end of the sort — but *which* end depends on your database engine. Postgres and MySQL disagree. Verify it, or specify `NULLS LAST` explicitly. I've watched this exact assumption put a wall of blanks at the top of a report.

One more distinction worth internalizing for interviews: `ORDER BY` *arranges* rows; `GROUP BY` *summarizes* them. They rhyme, but they do opposite things.

Understanding result ordering is a small idea with a big payoff — it's the difference between using a database and understanding one.

Full lesson (Lesson 4 of the series) in the comments.

SQL #DataEngineering

#SoftwareEngineering #LearnToCode

#Database

Twitter/X Thread

1/ Your SQL query returns rows in a different order every time you run it — and nothing is broken.

Here's why that happens, and how to fix it for good. 📖

2/ When you run a `SELECT`, the database hands back a stack of matching rows.

The part beginners never expect: it does NOT promise any particular order.

Without instructions, you get whatever was fastest for the database.

3/ Think of a deck of cards dumped on a table.

Run the query today, Ada's on top. Run it tomorrow, Ben's on top.

Nothing's wrong — the database just never agreed to sort anything.

4/ The fix: `ORDER BY`, then the column name.

```
SELECT name, score
FROM students
ORDER BY score DESC;
```

Now it's sorted the same way. Every. Single. Time.

5/ Two directions:

- `ASC` → small to big (1, 2, 3) • `DESC` → big to small (leaderboard)

`ASC` is the default. `DESC` is the one you have to ask for on purpose.

6/ One column isn't always enough.

```
ORDER BY class, name
```

SQL reads left to right: sort by class first, and only break ties with name.

List columns in order of importance.

7/ You can sort by more than plain columns:

- text (alphabetical)
- dates (oldest → newest)
- computed expressions

```
ORDER BY price * quantity DESC
```

The DB does the math, then sorts on the result.

8/ The NULL trap 🤖

NULLs (blanks) clump at one end when sorting — but WHICH end depends on your database. Postgres and MySQL disagree.

I once assumed bottom, got a wall of blanks at the top. Check before you trust it.

9/ Don't confuse these:

- `ORDER BY` → arranges the rows you have
- `GROUP BY` → collapses rows into summary buckets

`ORDER BY` sorts. `GROUP BY` summarizes. You can use both together.

10/ Sort in the database, not your app — it's built for it and can use indexes.

That's `ORDER BY`. Master it and your results always show up exactly the way you meant.

Full lesson 👉 {LINK}

Facebook Post

Ever run a database query, get a nicely ordered list, then run it again later and everything's in a totally different order? 🤖

Good news: nothing's broken! The database just never promised to sort your rows unless you ask it to. Think of it like dumping a deck of cards on a table — all the cards are there, but there's no real order until someone lines them up.

The magic word is `ORDER BY`. Add it, and your results come back sorted exactly how you want — smallest to biggest, A to Z, oldest to newest, every single time. There's even a sneaky gotcha with blank (NULL) values that trips up almost everyone.

I break it all down step by step in Lesson 4 of the series — beginner-friendly, no jargon. Come learn something today! 📌 {LINK}

Reddit Summary

Understanding why SQL returns rows in "random" order — and how ORDER BY actually works

A common early confusion: you run a `SELECT`, the rows look sorted, and you assume they'll always come back that way. They won't. A relational table is a set of rows, and a set has no guaranteed order. Without `ORDER BY`, the database returns rows in whatever order was cheapest — physical layout, index traversal, parallel execution — and that can differ between runs, versions, and servers.

Key points worth knowing:

- `ORDER BY column` imposes a repeatable order. `ASC` (ascending) is the default; `DESC` must be specified.
- Multiple columns are read left to right — the first is primary, the rest break ties. Order them by importance.
- You can sort by expressions, e.g. `ORDER BY price * quantity DESC`. The DB evaluates, then sorts.
- **NULL placement is engine-dependent.** Postgres sorts NULLs last in ASC; MySQL sorts them first. Use `NULLS FIRST/LAST` where supported instead of assuming.
- For pagination, always end `ORDER BY` with a unique column — otherwise ties resolve unpredictably and `LIMIT / OFFSET` can skip or repeat rows.
- `ORDER BY` (arranges rows) is not `GROUP BY` (collapses rows into summaries).

Generally prefer sorting in the database over app code — it can use indexes your app can't see. Sort in the app only when the rule can't be expressed in SQL.

GITHUB README

```
# SQL ORDER BY – Sorting Query Results Correctly

> Lesson 4 of the **Fun with Learning Technology** SQL series.
```

Problem

When you run a ``SELECT``, the database returns matching rows as a **result set**. Critically, **it does not guarantee any order** unless you ask for one. The same query can return rows in different orders across runs, engine versions, and servers – because the database returns them in whatever order was fastest.

The goal: take an unordered result set and arrange it into a specific, **repeatable** order.

Intuition

A relational table is a **set** of rows, and sets have no inherent order. Picture a deck of cards dumped on a table – the cards are all there, but there's no sequence until you hand the deck to someone with a sorting rule.

``ORDER BY`` is that rule. You give the database a comparison instruction, and it lines the rows up accordingly – every time.

Approach

1. **Single column:** ``ORDER BY score`` sorts ascending by default.
2. **Direction:** ``ASC`` (small → big, default) or ``DESC`` (big → small). ``DESC`` must be requested explicitly.
3. **Tie-breakers:** list multiple columns, read left to right, in order of importance: ``ORDER BY class, name``.
4. **Expressions:** sort by computed values: ``ORDER BY price * quantity DESC``.
5. **NULLs:** they clump at one end – **which** end is engine-dependent. Use ``NULLS FIRST`` / ``NULLS LAST`` where supported.

Python Solution

```
```python
import sqlite3

def build_db() -> sqlite3.Connection:
 conn = sqlite3.connect(":memory:")
 conn.execute(
 "CREATE TABLE students (id INTEGER, name TEXT, class TEXT, score INTEGER)"
)
 conn.executemany(
 "INSERT INTO students VALUES (?, ?, ?, ?)",
 [
 (1, "Ada", "B", 88),
 (2, "Ben", "A", 60),
 (3, "Chen", "A", 95),
 (4, "Dara", "B", 88),
 (5, "Evan", "A", 72),
],
)
 return conn
```

```
def top_scores(conn: sqlite3.Connection) -> list[tuple[str, int]]:
 """Return (name, score) sorted highest first; name breaks score ties."""
 cursor = conn.execute(
 "SELECT name, score FROM students ORDER BY score DESC, name ASC"
)
 return cursor.fetchall()

if __name__ == "__main__":
 result = top_scores(build_db())
 assert result == [
 ("Chen", 95), ("Ada", 88), ("Dara", 88), ("Evan", 72), ("Ben", 60),
], result
 print("Sorted:", result)
```

## Equivalent SQL

```
SELECT name, score
FROM students
ORDER BY score DESC, name ASC;
```

## Complexity

Metric	Value	Note
Time	$O(n \log n)$	Comparison sort; $O(1)$ extra if a matching index exists
Space	$O(n)$	Rows materialized for sorting; may spill to disk

## Common Pitfalls

- Assuming rows are ordered without `ORDER BY`.
- Forgetting `DESC` and getting ascending results.
- Assuming NULL placement (engine-dependent).
- Confusing `ORDER BY` (sorts) with `GROUP BY` (summarizes).
- No unique tie-breaker in paginated queries.

## Video

 Watch the full lesson: {LINK}

## Article

 Full written deep-dive with diagrams, dry runs, and interview prep: {LINK}

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Part of the **Fun with Learning Technology** series. ``

## Quick Review

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### When do you reach for ORDER BY in SQL?

When result order matters — reports, rankings, sequential processing. Without it, row order is unspecified and can change between runs.

### Default sort direction, and how to reverse it?

ASC (ascending) is default. Use DESC per column for descending. Directions are set individually, e.g. ORDER BY a ASC, b DESC.

### How does ORDER BY handle multiple columns?

Sorts by the first column, then breaks ties with the next, and so on. Order of columns in the clause defines precedence.

### Common bug: where do NULLs land when sorting?

NULL ordering is engine-dependent (often first in ASC on some DBs, last on others). Use NULLS FIRST/LAST or COALESCE to force it.

### ORDER BY vs GROUP BY — what's the difference?

GROUP BY collapses rows into aggregate buckets; ORDER BY only arranges output rows. Different jobs — sorting doesn't group, grouping doesn't sort.

### Follow-up: sort in the database or the app?

Prefer the database — it's optimized, can use indexes, and cuts data transferred. Sort in-app only for tiny, already-fetched sets.

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sql Lesson 4: Sorting with ORDER BY (The ORDER BY clause provides the necessary mechanism to arrange query results in a specific sequence based on defined column values. Without this clause, SQL databases return rows in an unpredictable order,

making data consumption inconsistent for users and downstream applications. You should reach for ORDER BY whenever the logical presentation of data is required for reporting, analysis, or sequential processing.) — free Python cheat sheet